

- 4 (a) State the principle of superposition.

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.....

..... [2]

- (b) An electromagnetic wave of wavelength 0.026 m in free space is incident normally on an aluminium sheet, as shown in Fig. 4.1.

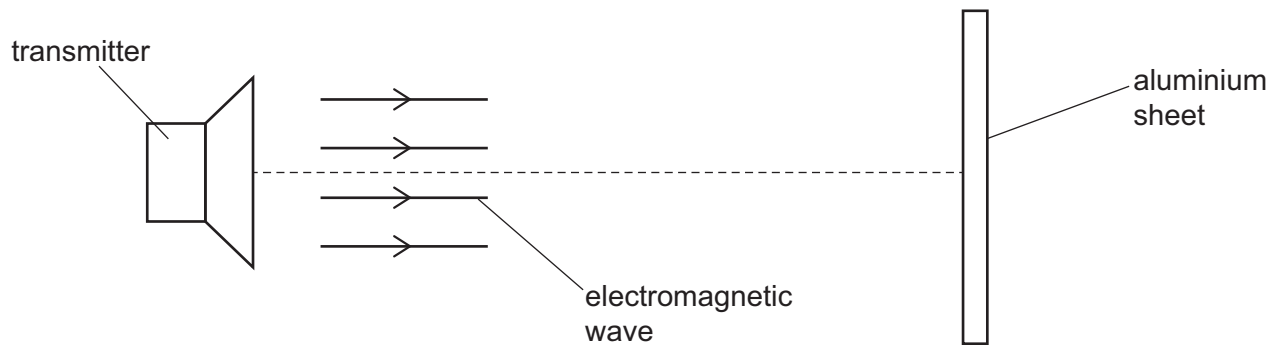


Fig. 4.1

The wave reflects at the aluminium sheet and a stationary wave is formed in the region between the transmitter and the sheet.

- (i) Explain how the stationary wave, including its nodes and antinodes, is formed.

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.....

..... [3]

- (ii) Calculate the frequency of the electromagnetic wave.

frequency = Hz [2]

- (iii) State the principal region of the electromagnetic spectrum to which the wave belongs.

..... [1]



(iv) Determine the distance between a node and an adjacent antinode.

distance = m [1]

[Total: 9]

